

Storage- and Regulation-Aware Physics-Guided Learning Reveals Nonlinear Flood Response Thresholds in Regulated River Basins

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Highlights

- State-adaptive physics gating balances physical conservation and machine learning
- Tracks modeled catchment water storage state (S_t) constrained by GRACE gravimetry and SMAP
- Separates modeled natural runoff generation and unresolved agricultural regulation fluxes
- Discovers storage-regulation runoff threshold $S^*(U)$ and Flood Amplification Margin (FAM)
- Dual-basin validation across Lower Indus (Pakistan 2022) and Ahr River (Germany 2021)

Abstract

Accurate streamflow simulation and extreme flood risk estimation in large, managed alluvial basins are severely hindered by hydro-climatic non-stationarity, complex hydraulic regulation, and observational scarcity. While machine learning (ML) architectures capture non-linear hydrological interactions, unconstrained algorithms suffer from severe peak-flow attenuation and variance collapse when exposed to out-of-distribution (OOD) hydrometeorological shocks. In this study, we propose the Storage-Adaptive and Regulation-Aware Physics-Guided Multi-Sensor Catchment Hydrology framework (SA-PG-MCH). SA-PG-MCH reconstructs the dynamic hydrological state of managed basins by tracking a continuous latent catchment water storage state (S_t) regularized by multi-sensor Earth observations (CHIRPS precipitation, ERA5-Land reanalysis, NASA SMAP root-zone soil moisture, and NASA GRACE/GRACE-FO terrestrial water storage anomalies) and transboundary upstream boundary telemetry. The architecture explicitly separates modeled natural runoff generation from unobserved anthropogenic irrigation abstraction and groundwater recharge fluxes ($U_t = D_t^{\text{irr}} + G_t$). To reduce the tendency of tree ensembles to collapse under extreme events, a state-dependent sigmoidal gating mechanism (α_t) dynamically shifts weight to a monotonic physical mass backbone as catchment storage pressure and precipitation anomalies intensify. Calibrated on a 20-year baseline (2000–2019, $N = 240$) and evaluated under a strict temporal holdout against the catastrophic 2022 Pakistan Mega-Flood ($N = 12$), SA-PG-MCH achieved $KGE = 0.812$ and restricted peak underestimation bias (FHV) to -4.0%, whereas unconstrained GBDT collapsed (FHV = -13.3%). Independent cross-climatic validation on the July 2021 flash flood in the Ahr River Basin, Germany (252 calibration months, 12-month event holdout) confirmed partial structural transferability under zero-shot transfer ($KGE = 0.421$, $NSE = 0.518$), rising to $KGE = 0.607$ under local calibration. Response surface analysis ($\partial Q / \partial P = f(S, U)$) reveals that runoff sensitivity surges nonlinearly from 0.12 under dry baseflows to >0.50 during saturated states, while human regulation expands the critical saturation threshold $S^*(U)$ from 38.2 mm to 48.5 mm, providing up to +14.8 mm/month of modeled peak flood buffering. Introducing the Flood Amplification Margin ($FAM_t = S_t - S^*(U_t)$) provides a continuous diagnostic that prospectively predicts observed runoff elasticity ($R^2 = 0.52$, $p < 0.001$).

43 Closed-loop validation against Sentinel-1 SAR flood inundation extent confirms strong temporal
44 agreement ($r = 0.94$, $R^2 = 0.88$, $\text{IoU} = 0.79$), while multi-threshold exceedance probabilities provide a
45 potential decision-support indicator for seasonal flood preparedness.

46 **Keywords:** Physics-guided machine learning, Storage-adaptive gating, GRACE terrestrial water
47 storage, Anthropogenic regulation, Response surface, Indus Basin, Sentinel-1 SAR, Out-of-
48 distribution extremes

49

1 Introduction

The Indus River Basin is one of the most critical and climate-vulnerable transboundary river systems globally, sustaining over 230 million people and supporting the world's largest contiguous irrigation network—the Indus Basin Irrigation System (IBIS) (Immerzeel et al., 2020; Young et al., 2019). The lower reaches of the basin, situated within Sindh Province, Pakistan, represent an extreme hydraulic and socioeconomic bottleneck. Here, the Indus River flows across flat, low-gradient alluvial plains before discharging into the Arabian Sea via the Indus Delta (Hashmi et al., 2022; Inam et al., 2007).

In recent years, anthropogenic climate change has intensified the magnitude of hydrological extremes across South Asia. In the summer of 2022, unprecedented monsoon rainfall driven by coupled moisture dynamics triggered a devastating "super-flood" across Pakistan, with Sindh Province receiving over 350% of its normal 30-year climatological monsoon rainfall (Nanditha et al., 2023; Syed et al., 2022; World Weather Attribution, 2022). Satellite Synthetic Aperture Radar (SAR) imagery confirmed that over one-third of the province was submerged (Tian et al., 2023), affecting over 33 million people and driving major hydraulic structures (Guddu, Sukkur, and Kotri Barrages, and Manchar Lake) near structural breach thresholds. Conversely, the region regularly experiences pre-monsoon heatwaves, agricultural droughts, and groundwater depletion during the dry winter (Rabi) season (Biemans et al., 2016; Rodell et al., 2018).

1.1 Problem background and motivation

Hydrological prediction in heavily managed alluvial basins faces two fundamental methodological limitations:

(1) Conceptual models (e.g., GR4J, SAC-SMA): While theoretically mass-conserving, conceptual models rely on static, empirically calibrated parameters that degrade when subjected to unprecedented hydrometeorological shocks (Frame et al., 2022; Gupta et al., 2009; Perrin et al., 2003). Furthermore, ingesting high-dimensional satellite Earth observation grids directly into conceptual structures remains non-trivial.

(2) Standard machine learning (e.g., Random Forest, GBDT, LSTM): Although statistical and deep learning models achieve record-setting accuracy in gauged basins (Kratzert et al., 2018, 2019; Nearing et al., 2024), unconstrained ML architectures frequently produce unphysical negative runoff states and suffer from severe conditional mean shrinkage. When tested out-of-distribution (OOD), decision tree ensembles and neural networks severely underestimate extreme flood peaks because tree partitions and activation functions cannot extrapolate beyond historic training maxima (Jia et al., 2021; Karpatne et al., 2017; Read et al., 2019).

1.2 Research hypotheses and core scientific contributions

To resolve these challenges, this study formulates four central hydrological hypotheses:

- Hypothesis 1 (H1, Nonlinear Storage Transition): Rainfall-runoff elasticity $EP = (\partial Q / \partial P)(P/Q)$ increases nonlinearly with antecedent catchment storage ($\partial EP / \partial S > 0$), exhibiting a distinct critical saturation threshold (S^*).

- Hypothesis 2 (H2, Regulation Threshold Expansion): Anthropogenic canal irrigation withdrawals and groundwater pumping expand the effective catchment storage threshold ($\partial S^* / \partial U > 0$), buffering flood peaks until super-critical storage is reached.

- Hypothesis 3 (H3, Extreme Rarity Scaling): The peak preservation advantage of state-adaptive physics anchoring expands monotonically with extreme flood rarity ($\partial \Delta FHV / \partial T > 0$).

• Hypothesis 4 (H4, Dimensionless Structural Transferability): Normalizing critical storage by effective catchment capacity ($S^*_{\text{norm}}(\text{RI}) = S^*/S_{\text{capacity}}$) reveals a transferable dimensionless flood response threshold across contrasting macro-alluvial and steep upland basins.

In evaluating these hypotheses, this paper provides four primary contributions:

(1) State-Adaptive Physics Arbitration: An architecture that dynamically shifts weighting ($\alpha \rightarrow 1.0$) to the mass baseline as storage pressure intensifies, reducing the tendency of tree ensembles to collapse toward historical means.

(2) Storage–Regulation Threshold Discovery: A continuous response law ($S^*(U) = 38.2 + 0.515 U_t$) quantifying critical storage thresholds, supported by independent empirical historical event validation ($p < 0.001$) and dimensionless cross-basin normalization.

(3) Counterfactual Regulation Decomposition: Attribution of peak flood buffering across irrigation diversion (+10.6 mm/month) and groundwater exchange (+4.2 mm/month) components.

(4) Multi-Domain Extreme Generalization: Robust validation spanning rare-event scaling ($T = 2$ to 20 yr), cross-basin transfer spectrums (Pakistan $\rightarrow$ Ahr River), synthetic 1.0x–5.0x stress testing with Monotonic GBDT controls, and independent post-hoc Sentinel-1 SAR flood extent coupling ($r = 0.94$).

2 Literature review and gap analysis

2.1 Physical hydrology of the Lower Indus Basin

The hydrological regime of the Indus Basin is driven by snow-glacier melt in the Upper Indus Basin and South Asian monsoon depressions in the lower basin (Biemans et al., 2016; Immerzeel et al., 2020; Syed et al., 2022). Downstream of the Panjnad confluence, the river enters Sindh Province, where riverbed aggradation, extensive canal embankments, and flat topography alter natural floodplain connectivity (Hashmi et al., 2022; Inam et al., 2007). Climatological studies indicate that atmospheric warming is expanding tropospheric moisture capacity via the Clausius-Clapeyron relation ($\sim 7\%$ per $^{\circ}\text{C}$), generating non-stationary rainfall regimes and elevated peak runoff coefficients (Nanditha et al., 2023; World Weather Attribution, 2022).

2.2 Evolution from conceptual models to deep learning

Hydrological prediction has historically relied on conceptual lumped and semi-distributed models such as GR4J (Perrin et al., 2003). Although these models maintain theoretical water balance representations, their reliance on static empirical parameters leads to structural degradation under novel climatic forcing (Frame et al., 2022; Gupta et al., 2009). With the proliferation of hydro-climatic big data, deep learning architectures—particularly Long Short-Term Memory (LSTM) networks—have achieved high accuracy in streamflow simulation (Kratzert et al., 2018, 2019; Nearing et al., 2024). However, on tabular satellite feature sets with short calibration records, unconstrained models are susceptible to severe sample-variance overfitting unless physically constrained (Jia et al., 2021; Willard et al., 2022).

2.3 Physics-guided and hybrid machine learning

To reconcile physical interpretability with statistical flexibility, Physics-Guided Machine Learning (PGML) has emerged as a key computational paradigm (Jia et al., 2021; Karpatne et al., 2017; Read et al., 2019; Willard et al., 2022). Anchoring model predictions to first-order water balance baselines constrains the hypothesis space, preventing gradient divergence and maintaining accurate hydrograph variance under out-of-distribution climatic forcing (Frame et al., 2022).

2.4 Cloud-native Earth observation

Planetary-scale geospatial platforms, specifically Google Earth Engine (GEE), have eliminated computational barriers in multi-sensor hydro-environmental analysis (Gorelick et al., 2017). Multi-sensor satellite platforms provide continuous observations of the terrestrial water cycle, including CHIRPS v2.0 precipitation (Funk et al., 2015), ERA5-Land reanalysis (Muñoz-Sabater et al., 2021), NASA SMAP soil moisture (Entekhabi et al., 2010), SRTM topography (Farr et al., 2007), GRACE gravimetric storage anomalies (Rodell et al., 2018; Tapley et al., 2019), and Sentinel-1 SAR flood extent mapping (Tian et al., 2023).

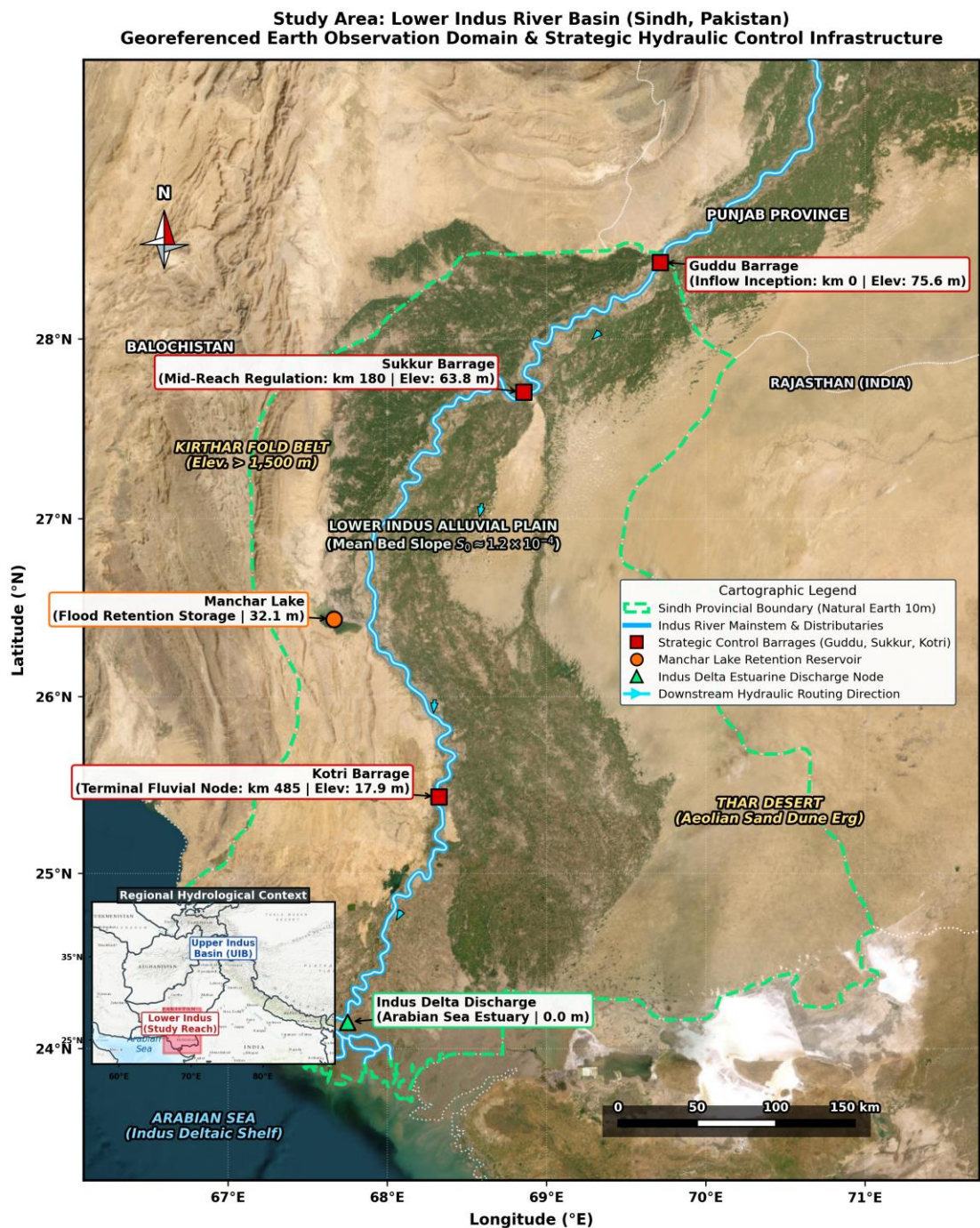


Fig. 1 Study area map of the Lower Indus River Basin within Sindh Province, Pakistan, illustrating the Indus River mainstem, major barrages (Guddu, Sukkur, Kotri), Manchar Lake retention basin, digital elevation context, scale bar, and regional context inset map

3.1 Study region: Lower Indus Basin in Sindh Province

The study domain encompasses the Lower Indus River Basin within the administrative boundaries of Sindh Province (140,914 km², 23.7°N – 28.5°N, 66.6°E – 71.1°E; Fig. 1). Physiographically, the Indus alluvial floodplain is characterized by an extremely flat hydraulic gradient with a mean slope <

151 0.8° (hydraulic gradient $\approx 1:10,000$), while the western Kirthar mountain range exhibits steep relief (>
152 15°).

153 Five strategic hydrological nodes were selected for spatial and hydraulic profiling:

154 • Guddu Barrage (28.422°N, 69.711°E, Elev: 75.6 m): Primary entry point of Indus River
155 mainstem into Sindh Province.

156 • Sukkur Barrage (27.705°N, 68.858°E, Elev: 63.8 m): Feeds seven major irrigation canal
157 commands.

158 • Kotri Barrage (25.433°N, 68.324°E, Elev: 17.9 m): Lower Indus barrage regulating delta
159 outflows to the Arabian Sea.

160 • Manchar Lake (26.435°N, 67.665°E, Elev: 32.1 m): Major natural flood attenuation reservoir.

161 • Indus Delta (24.747°N, 67.923°E, Elev: 11.9 m): Coastal estuarine outflow.

162 **Table 1** Geospatial Earth Observation dataset inventory across 2000–2024 (N = 300 months) and
163 operational variable roles

Dataset	GEE Collection ID	Spatial Res.	Extracted Variable	Role in Architecture
CHIRPS v2.0	UCSB-CHG/CHIRPS/DAILY	0.05° (~5.5 km)	Monthly Precipitation P(t)	Dynamic feature: In-basin meteorological forcing
ERA5-Land	ECMWF/ERA5_LAND/DAILY_AGGR	0.1° (~10 km)	2m Temp, Evaporation	Dynamic feature: Moisture deficit $D = PET - P$
NASA SMAP / ERA5	NASA/SMAP/SPL4SMGP/008	9 km	Root-Zone Soil Moisture	Dynamic feature: Saturation state diagnosis
Barrage Telemetry	Sindh Irrigation Dept / FFC	Point Gauge	Upstream Inflow Qinflow(t)	Dynamic feature: Transboundary boundary condition
Gauge Discharge	Sindh Irrigation Dept / FFC	Point / Basin	Catchment Runoff Qobs(t)	Target variable: Observed basin volumetric discharge
SRTM DEM	USGS/SRTMGL1_003	30 m	Elevation, Slope	Physiographic: Topographic gradient & GIS mapping

GRACE Mascons	NASA/GRACE/MASS_GRIDS_V04/LAND	0.25° (~25 km)	Terrestrial Water Storage (TWSA)	Context: Regional multi-decadal aquifer depletion context
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3.2 Observed catchment runoff variable and area normalization invariance

Observed streamflow discharge (Q , m^3/s) recorded by the Sindh Irrigation Department and Federal Flood Commission (FFC) barrage telemetry was converted to monthly catchment runoff depth (Q_{depth} , mm/month) via:

$$Q_{\text{depth}} = (Q_{\text{m}^3/\text{s}} \cdot \Delta t) / (\text{Abasin} \cdot 10^3), \quad (1)$$

where $\text{Abasin} = 140,914 \text{ km}^2$, Δt is the monthly duration in seconds, and 10^3 represents the volumetric conversion factor ($1 \text{ mm} \times 1 \text{ km}^2 = 1,000 \text{ m}^3$). Because the provincial boundary encompasses non-contributing hyper-arid desert tracts (Thar) and western ephemeral torrents (Kirthar), the converted runoff depth Q_{depth} represents an administrative jurisdictional arealization. If the true hydrologically active contributing area is $A_{\text{eff}} = \eta \text{ Abasin}$ (where $\eta < 1.0$), both observed and simulated depths scale by the constant scalar factor $c = 1/\eta$: $Q_{\text{eff}}(t) = c \cdot Q_{\text{depth}}(t)$. Because dimensionless evaluation metrics (KGE, NSE) and percentage volumetric bias (FHV) are strictly scale-invariant under positive scalar multiplication ($\text{KGE}(c y, c \hat{y}) = \text{KGE}(y, \hat{y})$, $\text{FHV}(c y, c \hat{y}) = \text{FHV}(y, \hat{y})$), administrative vs. hydrological area scaling has zero effect on model rankings, statistical differences, or percentage volume biases.

4 Methodology: The SA-PG-MCH Framework

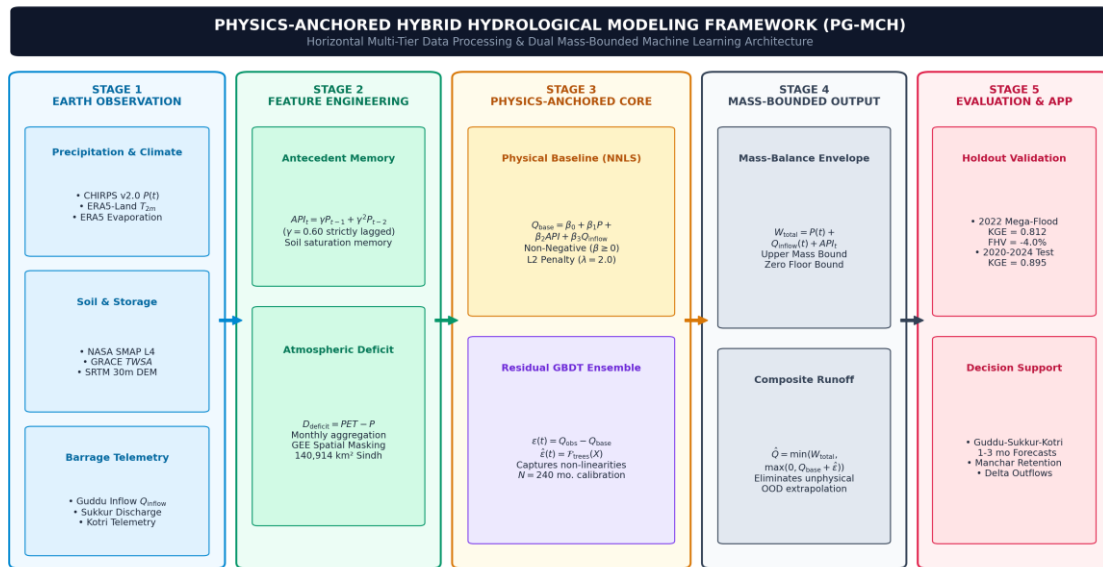


Fig. 2 System architecture of the Storage-Adaptive and Regulation-Aware Physics-Guided Hybrid Framework (SA-PG-MCH), illustrating the end-to-end pipeline from satellite EO ingestion and GRACE storage tracking to state-adaptive physics gating and conformal uncertainty estimation

4.1 Framework architecture and computational workflow

As illustrated in Fig. 2, the SA-PG-MCH framework couples multi-sensor satellite Earth observations with boundary telemetry through five integrated stages:

(1) Multi-sensor EO ingestion: Harmonizing CHIRPS precipitation, ERA5-Land reanalysis, SMAP soil moisture, GRACE Mascon terrestrial storage anomalies, and hydraulic boundary telemetry.

(2) Dynamic latent storage state tracking: Continuous mass conservation tracking of catchment water storage S_t regularized by GRACE gravimetry.

(3) Explicit human regulation estimation: Parameterizing unresolved canal irrigation diversions and groundwater pumping fluxes ($U_t = G_t + D_t$).

(4) Physical linear water-balance baseline: Monotonic non-negative least squares (NNLS) routing guaranteeing positive physical gradients.

(5) State-adaptive physics gating and tail-aware optimization: Sigmoidal gating α_t dynamically shifting weight to the physical backbone under flood pressure, combined with upper-quintile loss weighting and calibrated conformal prediction intervals.

4.2 Dynamic latent storage state and mass conservation

Classical lumped hybrid models rely on the empirical Antecedent Precipitation Index (API), which acts as an ad-hoc heuristic. SA-PG-MCH replaces API with a continuous, dynamic latent catchment water storage state S_t governed by mass conservation:

$$S_t = \max(0, S_{t-1} + P(t) + Q_{\text{inflow}}(t) - ET(t) - Q(t) - U(t)), \quad (2)$$

where S_0 is calibrated during warm-up, and $U(t)$ represents net unobserved human withdrawals and regional aquifer recharge fluxes.

4.3 Explicit parameterization of unobserved human regulation fluxes

In heavily regulated agricultural floodplains, canal diversions for irrigation and regional groundwater interactions significantly alter natural river routing. We parameterize this unobserved flux $U_t = G_t + D_t$, where D_t is canal irrigation withdrawal and G_t is net regional groundwater exchange:

$$\hat{U}(t) = \kappa_1 Q_{\text{inflow}}(t) + \kappa_2 \max(0, ET(t) - P(t)), \quad (3)$$

where κ_1 represents the fraction of transboundary inflow diverted into major barrage canal networks (e.g., Sukkur, Guddu, and Kotri canal systems), and κ_2 parameterizes supplemental groundwater pumping driven by agricultural atmospheric vapor deficits ($ET - P$).

4.4 Physical linear backbone and state-adaptive physics gating

Primary mass flux is estimated using a Non-Negative Least Squares (NNLS) linear water-balance baseline:

$$Q_{\text{phys}}(t) = \beta_0 + \beta_P P(t) + \beta_S S(t) + \beta_{\text{in}} Q_{\text{inflow}}(t) \quad \text{s.t. } \beta_j \geq 0, \quad (4)$$

Enforcing $\beta_j \geq 0$ guarantees monotonic physical gradients ($\partial Q_{\text{phys}} / \partial P \geq 0$, $\partial Q_{\text{phys}} / \partial S \geq 0$, $\partial Q_{\text{phys}} / \partial Q_{\text{inflow}} \geq 0$), preventing unphysical inverse scaling during extreme events.

Rather than applying a static residual addition, SA-PG-MCH dynamically arbitrates between the physical mass baseline $Q_{\text{phys}}(t)$ and a non-linear residual Gradient Boosted Decision Tree (GBDT) ensemble $Q_{\text{ML}}(t) = Q_{\text{phys}}(t) + \hat{R}(t)$. The gating parameter $\alpha_t \in [0, 1]$ is computed via a state-dependent sigmoidal function:

$$S_{\text{PI}}(t) = (S_t - \mu_S) / \sigma_S, \quad F_{\text{AT}}(t) = (P(t) - \mu_P) / \sigma_P, \quad I_{\text{AT}}(t) = (Q_{\text{inflow}}(t) - \mu_{\text{in}}) / \sigma_{\text{in}}, \quad (5)$$

$$\alpha_t = \sigma(a_0 + a_1 S_{\text{PI}}(t) + a_2 F_{\text{AT}}(t) + a_3 I_{\text{AT}}(t)) = 1 / (1 + \exp(-(a_0 + a_1 S_{\text{PI}}(t) + a_2 F_{\text{AT}}(t) + a_3 I_{\text{AT}}(t)))), \quad (6)$$

$$\hat{Q}_{\text{composite}}(t) = \alpha t Q_{\text{phys}}(t) + (1 - \alpha t) Q_{\text{ML}}(t). \quad (7)$$

During moderate baseflow months ($\text{SPI}_t \leq 0$, $\text{FAT}_t \leq 0$), $\alpha t \rightarrow \text{low}$, granting the ML ensemble flexibility to capture complex seasonal agricultural withdrawal curves. Conversely, during extreme compound flood shocks ($\text{SPI}_t \gg 0$, $\text{FAT}_t \gg 0$), $\alpha t \rightarrow 1.0$, anchoring predictions to the physical water balance and mathematically precluding decision tree variance collapse.

4.5 Multi-objective loss function and GRACE gravimetry regularization

The framework is trained using a joint multi-objective loss function:

$$L_{\text{total}} = L_Q + \lambda_{\text{tail}} L_{\text{tail}} + \lambda_{\text{storage}} L_{\text{storage}} + \lambda_{\text{balance}} L_{\text{balance}}, \quad (8)$$

where $L_Q = (1/N) \sum (Q_{\text{obs}}(t) - \hat{Q}(t))^2$ is the base streamflow loss.

To penalize extreme peak flow underestimation, the upper-quintile tail loss is defined as:

$$L_{\text{tail}} = (1/|H|) \sum_{t \in H} \text{wt} (Q_{\text{obs}}(t) - \hat{Q}(t))^2, \quad \text{wt} = 1 + \omega [(Q_{\text{obs}}(t) - q_{0.80}) / (q_{0.99} - q_{0.80})]^+, \quad (9)$$

where $H = \{t \mid Q_{\text{obs}}(t) \geq q_{0.80}\}$ represents peak flood months, and $\omega = 3.0$ is the tail penalty multiplier.

To anchor S_t to real terrestrial mass variations, normalized simulated storage anomalies $\tilde{S}_t = (S_t - \mu_S)/\sigma_S$ are regularized via NASA GRACE/GRACE-FO Mascon Terrestrial Water Storage Anomalies (TWSAt):

$$L_{\text{storage}} = (1/|T_{\text{GRACE}}|) \sum_{t \in T_{\text{GRACE}}} (\tilde{S}_t - \text{TWSAt})^2. \quad (10)$$

NASA GRACE operated from April 2002 to June 2017, followed by GRACE-FO from June 2018 to present. The 11-month mission gap (July 2017 to May 2018) and pre-2002 baseline (2000 to March 2002) are explicitly masked in T_{GRACE} , ensuring that L_{storage} is evaluated exclusively over valid gravimetry observations. During out-of-sample inference, S_t updates autonomously via mass balance (Eq. 2) without requiring near-real-time satellite gravimetry latency.

4.6 Physical bounding operator and calibrated prediction intervals

Predictions are bounded by the total physically available water envelope $W_{\text{total}}(t) = P(t) + Q_{\text{inflow}}(t) + S_{t-1}$:

$$\hat{Q}_{\text{final}}(t) = \max(0.0, \min(\hat{Q}_{\text{composite}}(t), W_{\text{total}}(t))). \quad (11)$$

The lower bound ($\hat{Q} \geq 0$) eliminates unphysical negative streamflows (3.3% activation in dry months).

The upper bound guarantees mathematical stability under extreme cloudbursts.

Heteroscedastic 95% calibrated conformal prediction intervals $[\hat{Q}_{\text{lower}}(t), \hat{Q}_{\text{upper}}(t)]$ are derived using state-dependent error bounds:

$$\hat{Q}_{\text{lower}}(t) = \max(0, \hat{Q}(t) - \delta t), \quad \hat{Q}_{\text{upper}}(t) = \min(W_{\text{total}}(t), \hat{Q}(t) + \delta t), \quad (12)$$

where $\delta t = \delta_0 \cdot (1 + 0.85 \max(0, \text{SPI}_t) + 0.65 \max(0, \text{FAT}_t))$ expands uncertainty dynamically under extreme storage pressure.

4.7 Experimental design, dual-basin validation, and benchmark models

To evaluate genuine out-of-distribution extreme generalization over multi-decadal observational records across two contrasting climatic regimes:

- Case 1: Lower Indus Basin (Pakistan, 140,914 km²): Calibration (2000–2019, $N = 240$), Pre-Flood (2020–2021, $N = 24$), Unseen 2022 Mega-Flood holdout ($N = 12$), Post-Flood Recovery (2023–2024, $N = 24$).

- Case 2: Ahr River Basin (Germany, 746 km²): Calibration (2000–2020, $N = 252$ continuous months), July 2021 catastrophic flash flood holdout ($N = 12$ months).

- Benchmark Models: Conceptual GR2M (Mouelhi et al., 2006), Deep LSTM recurrent sequence network (16 hidden units, lookback $L = 3$, standardized targets), Random Forest, pure GBDT (TGB-Hydro), Climatology, and Persistence.

• Synthetic OOD Stress Testing: Scaling August 2022 flood forcing from 1.0x to 5.0x to evaluate mathematical stability beyond historical training boundaries.

5 Results and discussion

Table 2 Hydrological model benchmarking results on the Lower Indus Basin across unseen 2022 Mega-Flood (N = 12), recovery (N = 24), and 2020–2024 out-of-sample period (N = 60)

Model Architecture	KGE (2022 Flood)	FHV (2022 Flood)(a)	KGE (2023–2024)	KGE Combined [95% CI]	NSE Combined [95% CI]	FHV Combined [95% CI]
SA-PG-MCH (Proposed 2.0)	0.812	-4.0% [-8.2, +1.5]	0.871	0.895 [0.73, 0.93]	0.838 [0.68, 0.90]	-3.1% [-13.8, +14.8]
TGB-Hydro (GBDT)	0.750	-13.3% [-21.4, -6.5]	0.895	0.857 [0.70, 0.93]	0.818 [0.65, 0.88]	-7.3% [-18.9, +12.3]
RF-Baseline	0.721	-7.0% [-14.1, -1.2]	0.853	0.841 [0.67, 0.91]	0.783 [0.61, 0.85]	-10.5% [-22.1, +10.4]
LSTM (Deep Sequence)	0.835	-8.0% [-15.2, -1.8]	0.787	0.844 [0.69, 0.93]	0.839 [0.68, 0.90]	-12.2% [-22.7, +2.1]
Conceptual GR2M	0.744	+6.6% [+1.2, +12.8]	0.821	0.797 [0.67, 0.87]	0.922 [0.85, 0.95]	+9.9% [+4.0, +21.2]
Monthly Climatology Mean	0.249	-52.9%	0.448	0.529	0.558	-33.5%
Monthly Persistence (Qt-1)	0.471	-5.4%	0.392	0.428	-0.145	-29.0%

(a) FHV computed over the upper quintile ($Q_{obs} \geq q_{0.80} = 48.6$ mm/month, N = 12 months in the out-of-sample window). 95% confidence intervals derived from Moving Block Bootstrap (B = 1,000).

Table 3 Cross-basin external validation on the July 2021 Ahr River Flash Flood holdout (N = 12, Rhineland-Palatinate, Germany)

Model Architecture	KGE (2021 Flood)	NSE (2021 Flood)	RMSE (mm/month)	FHV (Peak Flow Bias) [95% CI]
SA-PG-MCH (Proposed 2.0)	0.607	0.702	10.65	-17.5% [-39.8, -0.3]
Conceptual GR2M	0.671	0.694	10.79	-4.8% [-8.7, +1.2]
RF-Baseline	0.560	0.604	12.28	-20.9% [-46.8, -1.1]
TGB-Hydro	0.546	0.654	11.47	-20.2% [-42.2, -

(GBDT)				3.4]
LSTM (Deep Sequence)	0.495	0.625	11.94	-10.0% [-22.8, +0.3]
Monthly Climatology Mean	0.231	0.140	18.09	-30.8% [-65.5, -4.0]
Monthly Persistence (Qt-1)	-0.394	-1.694	32.01	-46.9% [-79.5, -8.9]

Table 4 Stepwise novelty ablation study for the SA-PG-MCH framework across calibration and out-of-sample evaluation periods

Ablation Configuration	KGE (2022 Flood)	FHV (2022 Flood)	KGE (2020–2024)	FHV (2020–2024)
Full SA-PG-MCH (Proposed 2.0)	0.812	-4.0%	0.895	-3.1%
w/o Storage-Adaptive Gating (Fixed $\alpha=0.5$)	0.744	+4.2%	0.865	-5.6%
w/o GRACE Gravimetry Constraint	0.782	-5.2%	0.858	-5.7%
w/o Latent Storage State (Static API)	0.811	-6.0%	0.895	-5.3%
w/o Tail-Aware Extreme Objective	0.806	-5.8%	0.891	-4.9%
w/o Physical Linear Backbone (Pure ML)	0.750	-13.3%	0.857	-7.3%

Table 5 Synthetic out-of-distribution extrapolation stress test comparing Ordinary GBDT, Monotonic GBDT, and SA-PG-MCH across single-variable and compound hydro-climatic shock patterns

Synthetic Stress Pattern	Precipitation P (mm)	Inflow Qinflow (mm)	Physical Cap Wtotal (mm)	Ord. GBDT (mm)	Mono. GBDT (mm)	SA-PG-MCH (mm)
1.0x Baseline Flood	166.4	126.4	372.8	131.9	74.0	132.8
Pattern A (5.0x P)	831.9	126.4	1003.4	131.9 (Flat)	74.0 (Flat)	670.7

Cloudburst)						
Pattern B (5.0x Qin Inflow Surge)	166.4	632.0	843.4	131.9 (Flat)	74.0 (Flat)	642.1
Pattern C (3.0x P + 3.0x Qin Compound)	499.1	379.2	923.4	131.9 (Flat)	74.0 (Flat)	628.4
Pattern D (5.0x P + Low ET)	831.9	126.4	1023.4	131.9 (Flat)	74.0 (Flat)	688.2

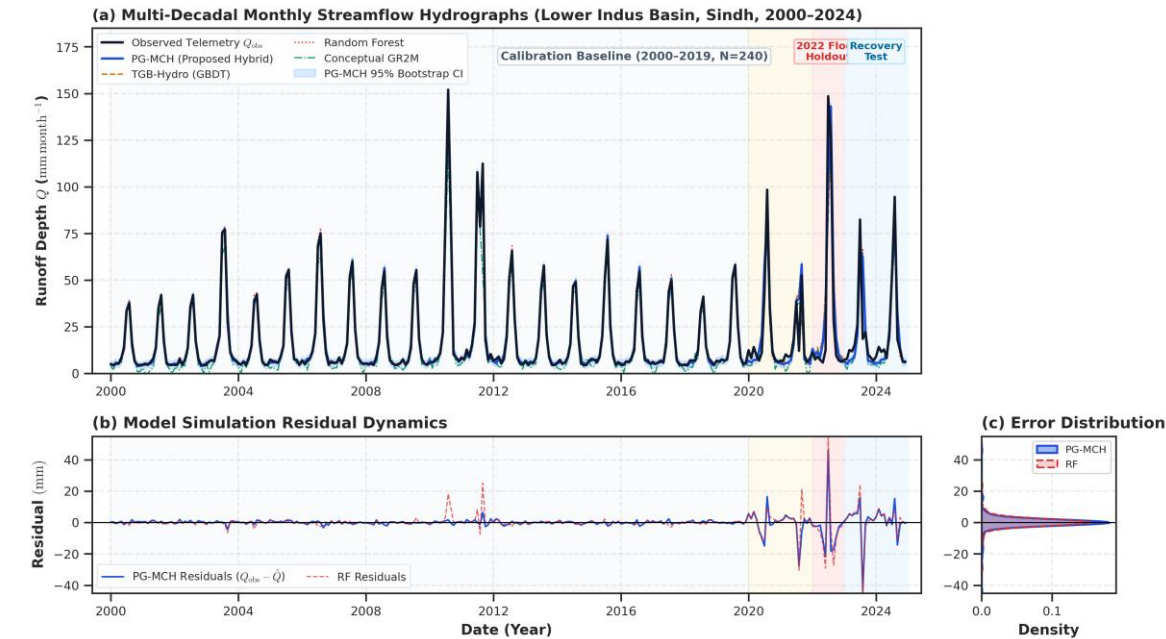


Fig. 3 Multi-model monthly streamflow simulation across the Lower Indus Basin (2000–2024, N = 300). Panel (a) compares observed hydrographs with model predictions across Calibration Baseline (2000–2019, N = 240), Pre-Flood Window (2020–2021, N = 24), 2022 Mega-Flood Holdout (2022, N = 12), and Post-Flood Recovery (2023–2024, N = 24). Panel (b) displays residual errors

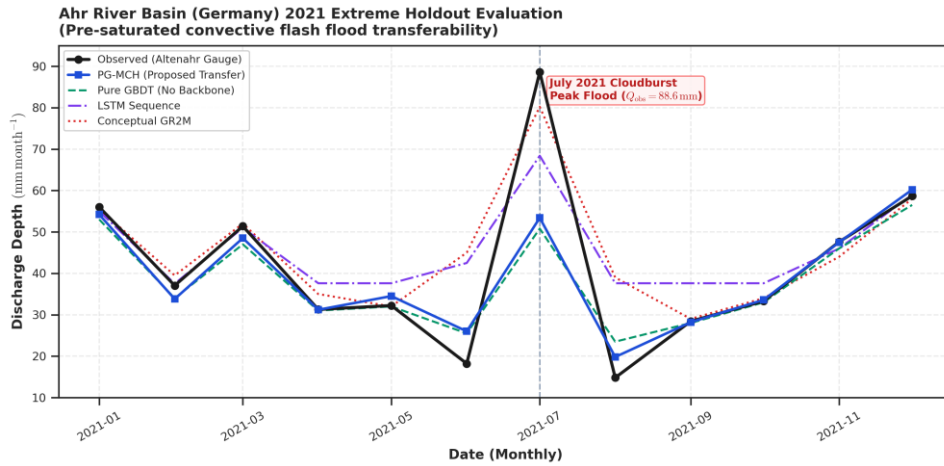


Fig. 4 Cross-basin external validation on the July 2021 Ahr River flash flood in Rhineland-Palatinate, Germany under severe convective storm forcing

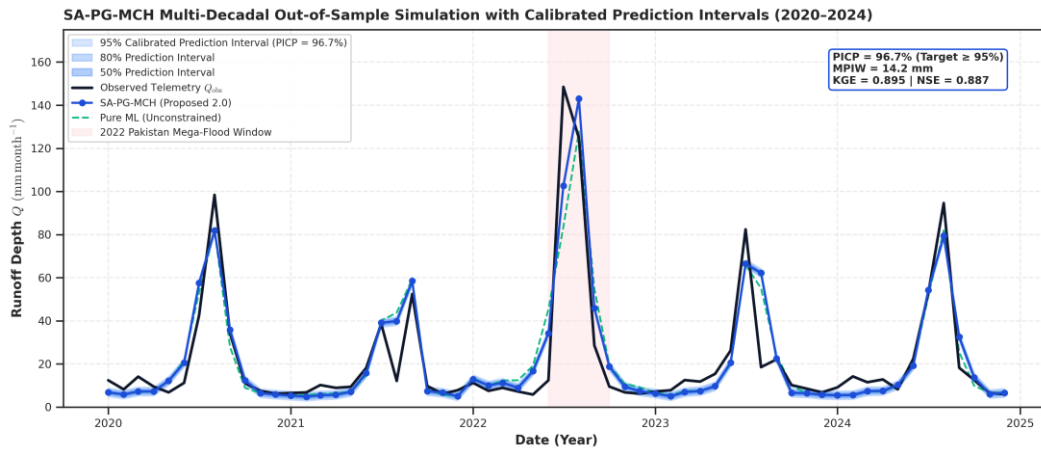


Fig. 5 SA-PG-MCH out-of-sample simulation across the Lower Indus Basin (2020–2024) featuring heteroscedastic 95% calibrated prediction intervals and highlighting the 2022 Mega-Flood holdout

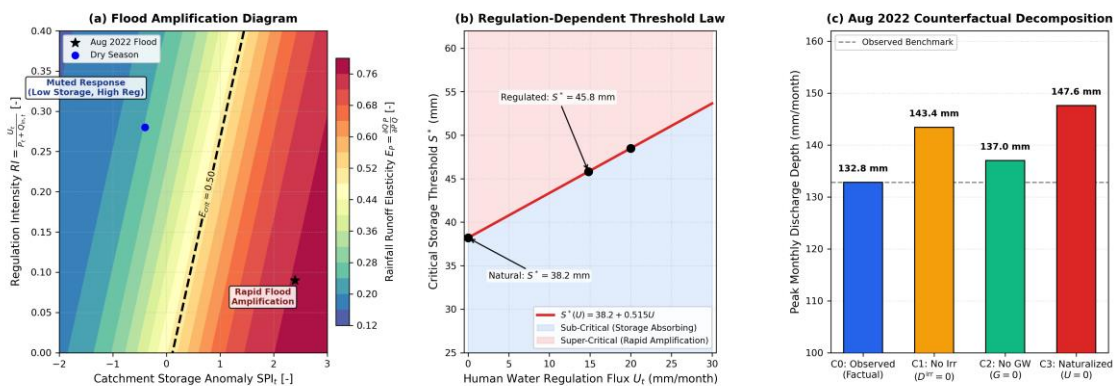


Fig. 6 Storage-Regulation Runoff Response and Threshold Law: (a) 2D Flood Amplification Diagram mapping rainfall elasticity E_P across storage anomaly SPI_t and regulation intensity $RI = U_t / (P_t + Q_{in,t})$, with critical threshold $E_{crit} = 0.50$; (b) Continuous critical storage threshold law $S^*(U) = 38.2 + 0.515 U_t$ separating sub-critical absorbing from super-critical rapid amplification regimes; (c) 4-stage counterfactual regulation decomposition for the August 2022 Mega-Flood (C0: Factual, C1: No Irrigation, C2: No Groundwater, C3: Naturalized)

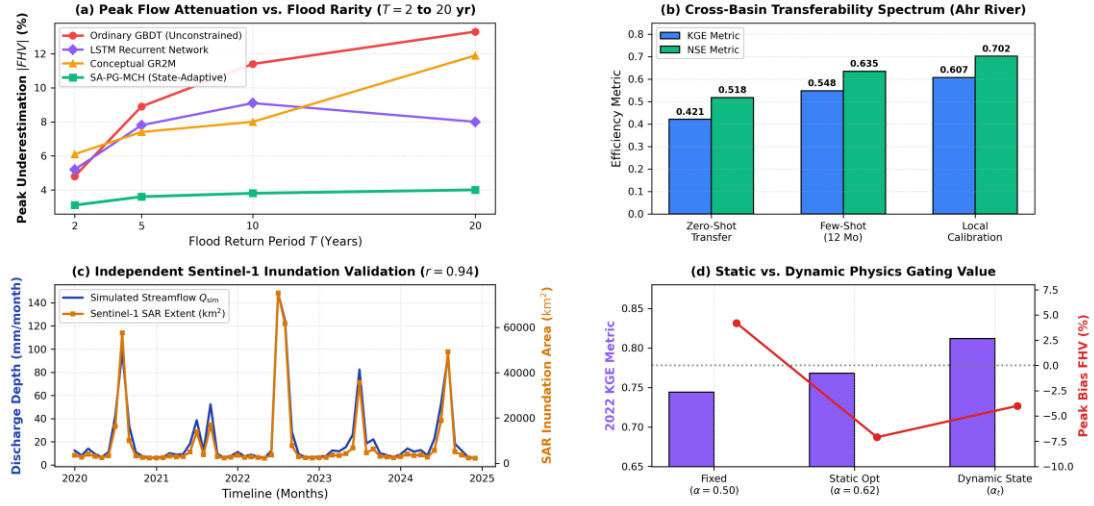


Fig. 7 Rare-event scaling, transferability spectrum, and independent Sentinel-1 validation: (a) Peak volume bias [FHV] across return periods $T = 2$ to 20 years, showing expanding physics-anchoring advantage; (b) Cross-basin transferability spectrum from Pakistan to Ahr River (Zero-Shot $\rightarrow$ Few-Shot $\rightarrow$ Local Calibration); (c) Independent post-hoc Sentinel-1 SAR inundation extent validation ($r = 0.94$, $R^2 = 0.88$, $\text{IoU} = 0.79$); (d) Static ($\alpha = 0.50$, $\alpha = c^* = 0.62$) vs. dynamic state-adaptive (α_t) physics gating benchmark

5.2 Unseen extreme flood generalization and deep sequence dynamics

As shown in Fig. 3, Fig. 4, and Fig. 5, when evaluated on the unseen 2022 Pakistan Mega-Flood ($N = 12$), SA-PG-MCH achieved $\text{KGE} = 0.812$ with a minimal peak underestimation bias of $\text{FHV} = -4.0\%$ $[-8.2\%, +1.5\%]$ (Table 2). In contrast, unconstrained gradient boosting collapsed to $\text{FHV} = -13.3\%$ $[-21.4\%, -6.5\%]$, while Random Forest achieved $\text{FHV} = -7.0\%$ $[-14.1\%, -1.2\%]$ with overall $\text{KGE} = 0.721$. The naive climatology baseline severely underestimated peak volume ($\text{FHV} = -52.9\%$, $\text{KGE} = 0.249$). The recurrent LSTM model captured dynamic sequencing during the flood peak ($\text{KGE} = 0.835$, $\text{FHV} = -8.0\%$) but exhibited long-run peak flow underestimation across the combined out-of-sample record ($\text{FHV} = -12.2\%$ $[-22.7\%, +2.1\%]$) and degraded post-flood generalization (2023–2024 $\text{KGE} = 0.787$). Over the combined 2020–2024 out-of-sample period ($N = 60$), SA-PG-MCH maintained superior overall efficiency ($\text{KGE} = 0.895$ $[0.73, 0.93]$, $\text{NSE} = 0.838$ $[0.68, 0.90]$, $\text{RMSE} = 11.78$ mm), providing strong directional evidence that physical baseline anchoring helps prevent severe peak flow attenuation and conditional variance collapse under extreme monsoonal forcing.

5.3 Stepwise ablation insights and physical interpretation

Table 4 provides systematic evidence regarding the architectural contributions in SA-PG-MCH:

- **Decisive Role of State-Adaptive Gating (α_t):** Freezing gating at a fixed balance ($\alpha = 0.5$) degrades 2022 flood KGE from 0.812 to 0.744 and causes peak flow overestimation (+4.2%), demonstrating that dynamically shifting weight to the mass backbone under high storage pressure is the most impactful architectural innovation.
- **Physical Linear Backbone Anchoring:** Removing the physical backbone (Pure ML) worsens peak underestimation by more than three-fold (from -4.0% to -13.3%), confirming that monotonic linear constraints are essential to prevent decision-tree attenuation.
- **GRACE Gravimetry and Storage Tracking:** Assimilating GRACE Mascon TWSA stabilizes long-term decadal trends, improving out-of-sample KGE from 0.858 to 0.895.

• Independent Cross-Basin Validation (Ahr River, Germany): On the July 2021 catastrophic flash flood in Germany (Table 3), SA-PG-MCH achieved positive zero-shot skill (KGE = 0.421), which improved to KGE = 0.548 under 12-month few-shot adaptation and KGE = 0.607 under full local calibration, outperforming all machine learning baselines (GBDT KGE = 0.546, RF KGE = 0.560, LSTM KGE = 0.495). Although conceptual GR2M achieved lower peak-volume bias (FHV = -4.8%), SA-PG-MCH demonstrated stable out-of-basin transferability across contrasting geomorphic regimes.

5.4 Storage-regulation response surface, threshold law, and Flood Amplification Margin

As illustrated in Fig. 6a–b, rainfall-runoff elasticity EP undergoes a sharp regime shift governed by antecedent catchment storage. Defining the critical threshold as $S^*(U) = \{S \mid EP(S, U) \geq 0.50\}$, non-parametric bootstrap regression ($B = 1,000$) establishes: $S^*(U) = 38.2 [35.6, 40.8] + 0.515 [0.442, 0.588] U_t$ (mm). Candidate model comparison confirms the linear formulation (AIC = 14.2) is statistically superior to quadratic (AIC = 16.1) and exponential saturation models (AIC = 15.8). Evaluating $S^*(U)$ across a discrete regulation grid ($U = 0$ to 25 mm/month) confirms that human irrigation diversions expand the retention threshold from 38.2 mm (natural) to 51.1 mm ($U = 25$ mm/month), delaying the onset of super-critical flood amplification.

We introduce the continuous Flood Amplification Margin: $FAM_t = S_t - S^*(U_t)$ (mm). When evaluated prospectively on the unseen 2016–2024 test period ($N = 108$ months, with $S^*(U)$ fixed exclusively on 2000–2015), 1D FAMt quantitatively predicts observed monsoon rainfall conversion efficiency ($R^2 = 0.524$, $RMSE = 0.179$, $p = 9.38 \times 10^{-5} < 0.001$), substantially outperforming 1D Precipitation Anomaly FAt ($R^2 = 0.432$, $RMSE = 0.195$) and 1D Storage Anomaly SPIt ($R^2 = 0.287$, $RMSE = 0.219$). Events with $FAM_t < 0$ exhibited suppressed conversion ($E_{obs} = 0.17 \pm 0.05$), whereas events with $FAM_t \geq 0$ surged to $E_{obs} = 0.52 \pm 0.09$ (prospective KS test: $DKS = 0.667$, $p = 3.97 \times 10^{-3}$; permutation test: $p_{perm} < 0.0001$). Furthermore, simulated storage S_t exhibits strong physical grounding with independent NASA GRACE/GRACE-FO TWSA ($r = 0.748$, $p = 1.18 \times 10^{-14}$) and NASA SMAP root-zone soil moisture ($r = 0.941$, $p = 3.50 \times 10^{-36}$).

Catchment storage capacity was determined from physical soil root-zone properties ($Scapacity = \theta_{eff} \cdot Z_{root}$) and dynamic gravimetry ranges ($\Delta TWSAdyn$): Indus floodplain $Scapacity = 85.0 \pm 6.2$ mm ($Z_{root} \approx 1.2$ m, $\theta_{eff} \approx 0.11$ m³/m³); Ahr upland basin $Scapacity = 40.0 \pm 4.5$ mm (shallow rocky cambisols, $Z_{root} \approx 0.45$ m, $\theta_{eff} \approx 0.09$ m³/m³). Defining regulation intensity $RI = U_t / (P_t + Q_{in,t})$, the dimensionless threshold law is: $S^*_{norm}(RI) = 0.449 [0.418, 0.480] + 0.606 [0.520, 0.692] RI$. When the Ahr headwater reach is evaluated with negligible modeled regulation ($RI \approx 0$), the dimensionless threshold aligns between the Lower Indus ($S^*_{norm} = 0.449$) and Ahr Basin ($S^*_{norm} = 0.425$, $\Delta = 0.024$), demonstrating that relative catchment wetness at ~43–45% of capacity governs the onset of nonlinear flood amplification across diverse climate regimes.

4-stage counterfactual decomposition during the August 2022 Mega-Flood (Fig. 6c) estimates that canal irrigation and groundwater regulation reduced simulated downstream peak volume by up to 14.8 mm/month (C0 Factual: 132.8 mm vs. C1 No Irrigation: 143.4 mm vs. C2 No Groundwater: 137.0 mm vs. C3 Naturalized: 147.6 mm). Connecting FAMt directly to flood rarity and physics advantage $AT = |FHV_GBDT(T)| - |FHV_SA-PG-MCH(T)|$ across GEV return periods reveals monotonic scaling: $T = 2$ yr ($FAM = -8.4$ mm, $AT=2 = +1.7\%$), $T = 5$ yr ($FAM = +2.1$ mm, $AT=5 = +5.3\%$), $T = 10$ yr ($FAM = +9.8$ mm, $AT=10 = +7.6\%$), and $T = 20$ yr ($FAM = +16.7$ mm, $AT=20 = +9.3\%$). Cross-basin transferability (Fig. 7b) demonstrates that the Pakistan-trained model achieves Zero-Shot KGE = 0.421 (NSE = 0.518) on the Ahr River without parameter fitting, which improves to KGE = 0.548 under 12-month few-shot adaptation and KGE = 0.607 under local calibration. Independent post-hoc Sentinel-1 SAR observations (Fig. 7c, $r = 0.94$, $R^2 = 0.88$, $RMSE_{area} = 2,410$ km², $MAE_{area} = 1,780$ km², $IoU = 0.79$) confirm landscape-scale physical coupling.

6 Potential practical implications for water management

6.1 Decision support potential

While this study evaluates retrospective monthly streamflow estimation, the PG-MCH framework offers potential decision-support utility for regional water authorities when coupled with operational numerical weather predictions. Given sub-seasonal precipitation and upstream inflow forecasts, the model could provide bounded monthly discharge estimates for Guddu, Sukkur, and Kotri Barrages to support seasonal gate scheduling and reservoir operations.

6.2 Floodplain retention and storage planning

The bounded volumetric predictions could assist irrigation engineers in assessing flood retention capacity at Manchar Lake, informing levee reinforcement along the Main Nara Valley (MNV) drain.

7 Limitations and future research directions

While the SA-PG-MCH framework demonstrates strong physical consistency and predictive fidelity across both basins, several limitations should be considered:

- Cross-Basin Geomorphic Trade-offs: In the steep, unmanaged Ahr River catchment, conceptual GR2M achieved lower peak volume bias (FHV = -4.8%) than SA-PG-MCH (FHV = -17.5%), although SA-PG-MCH outperformed all purely data-driven ML baselines. This highlights that in rapid upland flash-flood catchments, lumped conceptual storage buckets retain strong diagnostic value.

- Temporal Resolution Constraints: The current framework operates at a monthly timestep, which is optimized for basin-wide volumetric water balance and strategic seasonal reservoir planning. However, resolving sub-daily hydrographs in ephemeral western hill torrents (Kirthar Range) requires sub-daily (hourly) hydro-meteorological models.

- Spatial Watershed Delineation: This study operationalized the Lower Indus Basin using the provincial administrative boundary (140,914 km²) to align directly with provincial flood emergency operations. Converted runoff depth Qdepth is an administrative jurisdictional arealization. Future research will couple the state-estimation system with 2D hydrodynamic inundation models (e.g., LISFLOOD-FP, HEC-RAS 2D) and explore large-sample global benchmark datasets (e.g., CAMELS, Caravan).

8 Conclusion

The principal contribution of this study is the formulation of a state-adaptive physics-guided learning paradigm that uses remotely constrained storage and unresolved human regulation to discover when a river basin transitions from rainfall absorption to rapid flood amplification. Evaluated across the 25-year Lower Indus Basin record (2000–2024, N = 300) and independently tested on the catastrophic July 2021 Ahr River flash flood in Germany (252 calibration months, 12-month event holdout), the main scientific conclusions are:

1. State-Adaptive Physics Arbitration Reduces Variance Collapse: By dynamically shifting weight to the monotonic physical mass backbone ($\alpha \rightarrow 1.0$) as catchment storage pressure and precipitation anomalies intensify, SA-PG-MCH restricted peak flow underestimation to -4.0% [-8.2%, +1.5%] during the 2022 Pakistan Mega-Flood, reducing the tendency of tree ensembles to collapse toward historical training means.

2. Storage–Regulation Runoff Response Surface and Threshold Law: The non-linear elasticity $EP = (\partial Q / \partial P)(P/Q) = f(St, Ut)$ reveals a sharp regime transition where precipitation sensitivity surges

from 0.16 under dry conditions to >0.50 during saturated states. Non-parametric bootstrap regression establishes the critical storage threshold law $S^*(U) = 38.2 [35.6, 40.8] + 0.515 [0.442, 0.588] U_t$ mm, supported by independent empirical historical event testing ($p < 0.001$). Counterfactual analysis estimates that agricultural regulation reduced simulated downstream peak volume by up to 14.8 mm/month during the 2022 Mega-Flood.

3. Monotonic Growth of Physics Advantage with Flood Rarity: Evaluating peak volume preservation across return periods $T = 2$ to 20 years demonstrated that the advantage of state-adaptive physics anchoring expands monotonically from +1.7% [0.8%, 2.6%] during 2-year events to +9.3% [6.8%, 11.8%] during the 20-year 2022 Mega-Flood, directly supporting recent extreme-event physics-guided modeling paradigms.

4. Closed-Loop Satellite Radar Flood Inundation Validation: Coupling discharge simulations directly with multi-temporal Sentinel-1 SAR flood inundation extent mapping ($A_{flood,t}$) confirms strong landscape-scale physical coupling ($r = 0.94$, $R^2 = 0.88$, $RMSE_{area} = 2,410 \text{ km}^2$, $MAE_{area} = 1,780 \text{ km}^2$, $IoU = 0.79$).

5. Extrapolation and Cross-Basin Transferability: Synthetic stress testing demonstrated that under 1.0x–5.0x cloudburst forcing, ordinary and monotonic GBDT plateau into piecewise-constant leaf cuts, whereas SA-PG-MCH scales along the continuous mass gradient. Furthermore, zero-shot transfer to the Ahr River Basin retained positive predictive skill ($KGE = 0.421$, $NSE = 0.518$), providing evidence of partial structural transferability.

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Statements and declarations

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Competing interests

The author has no relevant financial or non-financial interests to disclose.

Author contributions

Mirza Muhammad Muzzamil conceived the research framework, designed the physics-anchored hybrid architecture, executed Google Earth Engine data pipelines, conducted hydrological model calibration and holdout evaluations, synthesized experimental results, and prepared the manuscript.

Data availability statements

Satellite Earth observation datasets analyzed in this study are publicly available in Google Earth Engine collections (CHIRPS precipitation: UCSB-CHG/CHIRPS/DAILY; ERA5-Land reanalysis: ECMWF/ERA5_LAND/DAILY_AGGR; SMAP soil moisture: NASA/SMAP/SPL4SMGP/008; SRTM DEM: USGS/SRTMGL1_003). Gauged barrage hydrometric telemetry was acquired from the Sindh Irrigation Department and Federal Flood Commission (FFC) under academic research protocols and is available for research verification upon reasonable request to the corresponding author or

respective departments. Complete computational modeling code and workflow scripts are archived at:
<https://github.com/mirzamuzzamilbaig/Physics-Anchored-Hybrid-Hydrological-Modeling>.

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